# Stationary Strings Theorem IV: Cosmic Fission, Vortex Fuel and the Singularity-Free Information Core

(Dimensional Fragmentation, Event Horizon Ring Fuel Mechanism, Resolution of the Information Paradox and Elimination of the $r = 0$ Singularity — Extended Complete Version)

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**Abstract**

As the fourth link of the *Stationary Strings Theorem* series, this study analyzes the cosmological and quantum-gravitational consequences of the "dimensional shear and ontological stripping" mechanism introduced in the third article [3]. The fundamental proposition of the study is that the separation of the two existential dimensions of matter — the **String (Frequency) Dimension** and the **Light (Photon) Dimension** — in the black hole vortex is a **Cosmic (Dimensional) Fission** process, just like the fragmentation of the atomic nucleus (nuclear fission); without this fragmentation, matter cannot pass into the stationary information realm ($\omega = 0$).

The fate of the two components emerging from this dimensional fragmentation eliminates two major unsolved problems of theoretical physics:

1. **Vortex Fuel and Resolution of the Information Paradox:** In agreement with the "annulus and oscillatory pulses ($\sigma = \pm 1$)" mechanism of the Navier–Stokes Millennium Problem solved in September 2026 [12], the **Light Dimension** stripped from the string turns into an intrinsic **Vortex Fuel** ($T_{\text{fuel}} = c_1 v_1 + c_2 v_2$) feeding the vortex at the event horizon and closing the singular momentum deficit. The specific energy lost by the string ($\Delta E_{\text{string}} \to 0$) is transferred to the vortex fuel encoded with the topological mode number $n$ ($\Phi_{\text{phase}} = n \ln X$). Thus, when black hole radiation and gravitational waves feed on this fuel, the $n$ information is carried back to the outer universe unitarily.
2. **Elimination of the $r = 0$ Singularity at the Center:** The **Stationary String**, stripped of vortex fuel below the event horizon with its vibration nulled

($\omega_n = 0$), completely loses its kinetic pressure ($P_{\text{kinetic}} = 0$) and produces only a negative vacuum-information pressure ($P_r = -\rho_{\text{information}} c^2$) originating from the topological information invariant ($\mathcal{I}_n = nc$). When this equation of state is solved in the Einstein field equations, the infinite-density singularity at $r = 0$ disappears, and a **Pure Information Core** of finite curvature forms at the center of the black hole.

In the final section, the **Quantum Echo** frequency predictions of the theory testable at the LIGO/Virgo/LISA gravitational wave observatories, and the **Acoustic/Superfluid Vortex Experiment** setup that can be built in the laboratory, are presented.

# 1 Introduction: Two Great Questions After Stripping

In the first three studies of the series [1, 2, 3], the cycle of the universe between the "Vibrating Energy Phase" ($\omega > 0$) and the "Stationary Information Phase" ($\omega = 0$) was established, and in the third study [3] it was proved that matter is a two-dimensional fabric composed of the **String Dimension** and the **Light Dimension**. Through the Polyakov string action integral and the September 2026 Navier–Stokes vortex singularity (*blow-up*) scaling laws [12], it was shown with the following fundamental conservation law how matter entering the black hole vortex is stripped in proper time ($\tau \to 0$) [3]:

$$\mathcal{I}_n = \omega_n(\tau) \cdot \mathcal{L}_{\text{light}}(\tau) = \left(\omega_{n,0}\, \tau^{+h}\right) \cdot \left(\ell_0\, \tau^{-h}\right) = nc = \text{Constant} \tag{1}$$

According to this equation, under vortex shear the light dimension of matter stretches to infinity ($\mathcal{L}_{\text{light}} \to \infty$) while the string dimension collapses to zero ($\omega_n \to 0$) in inverse proportion, and the fabric of matter splits in two, just as skin is flayed from a body [3].

At this point, two fundamental questions arise that will both complete the inner integrity of the *Stationary Strings Theorem* itself and resolve the two greatest crises of modern physics (the Black Hole Information Paradox and the $r = 0$ Central Singularity):

1. *If the string completely nulls its own inner vibration and specific energy ($E_{string,n} = \hbar\omega_n(\tau) \to 0$) as it descends to the event horizon, where does this vanishing specific energy go, as required by the law of conservation of energy? What function does that Light Dimension ($\mathcal{L}_{light} \to \infty$), flayed like skin from the body, serve inside the vortex?*

2. *How do the Stationary Strings, transformed into a pure information core ($\omega_n = 0$, $\mathcal{I}_n = nc$) stripped of their energy shell, alter the spacetime geometry below the event horizon and at the center of the black hole ($r \to 0$)?*

To answer these two questions, this fourth article proposes and mathematically proves the following fundamental hypothesis:

"The separation of the sheared String and Light dimensions in the black hole vortex is an existential fission, just like the fragmentation of the atom; without this fragmentation, matter cannot shed its energy load and pass into the information realm. The stripped light and specific-energy dimension does not vanish; it turns directly into an intrinsic **Vortex Fuel** rotating and sustaining the black hole vortex. The **Stationary String**, which leaves its specific energy as fuel to the vortex, forms a negative-pressure **Pure Information Core** at the center, eliminating the $r = 0$ singularity."

# 2 Cosmic Fission: The Obligatory Fragmentation Threshold of Transition into Information

## 2.1 From Nuclear Fission to Dimensional Fission

In nuclear physics, unless a heavy atomic nucleus fragments (fission), the binding energy holding the nucleus together cannot be liberated and the nucleus cannot transform into more stable subcomponents. Similarly, the material world we perceive is a composite nucleus in which the **String Dimension** ($\Psi_{\mathbf{string}}$) and the **Light Dimension** ($\Psi_{\mathbf{light}}$) are bound by a strong ontological tension [3].

In normal spacetime, matter cannot pass directly into the "Information Realm" (the dimension of perfect order) because of the kinetic vibration energy ($\hbar\omega_n > 0$) and entropy load it carries; for energy and vibration by their nature produce time and disorder (entropy) [1, 2]. For matter to transform into pure information ($\omega_n = 0$), this dynamical energy shell must be forcibly torn from the core.

Here the hyperbolic shear in the black hole vortex ($\mathrm{Tr}(\Sigma_{\mu\nu}) = 0$) pulls these two dimensions of matter toward opposite poles ($\tau^{-h} \to \infty$ and $\tau^{+h} \to 0$), performing the **Dimensional Fission (Cosmic Fragmentation)**:

$$\underbrace{\text{Matter Core } (\Psi_{\text{string}} \otimes \Psi_{\text{light}})}_{\text{Bound Matter Fabric}} \xrightarrow{\text{Cosmic Fission (Vortex Shear)}} \underbrace{\Delta E_{\text{fuel}}(\mathcal{L}_{\text{light}} \to \infty)}_{\text{Vortex Fuel (Light Ring)}} + \underbrace{\mathcal{I}_n(\omega_n = 0)}_{\text{Pure Information (Stationary String)}} \tag{2}$$

This equation explains why dimensional shear is not an "annihilation" but an obligatory **purification and fragmentation threshold** for transition into information: without fragmentation of matter, information cannot be liberated.

# 3 Vortex Fuel Mechanism: The Stripped Light Dimension Feeding the Vortex

## 3.1 The "Annulus and Pulses" Proof in the Navier–Stokes Solution

One of the deepest mechanisms of the *Finite Time Blowup for Navier–Stokes* solution of September 2026 [12] examined in the third article [3] is presented in **Section 2.2 (Annulus and pulses)** [12, pp. 5–6] and **Section 3 (Admissible stress cone condition)** [12, p. 10] of the article.

In the Navier–Stokes vortex singularity, while the kinetic energy of the inner core (inner core) collapsing toward the center goes to zero ($E_{\text{core}} \asymp \tau^{1/2-3h} \to 0$ [12, p. 4]), a huge momentum and force imbalance emerges in the **annulus region** surrounding the outer boundary of the core. For the singularity to continue by itself without an infinite external force, the fluid must produce a **fuel (intrinsic momentum flux)** from its own inner motion. In the solution this mechanism is proved as follows [12, pp. 5–6]:

1. **Energy Extraction from Background Shear:** Spatially oscillatory ring pulses from two different families ($\sigma = +1$ and $\sigma = -1$) located in the annulus region grow exponentially by extracting energy directly from the shear stress of the vortex (*"The pulses grow by extracting energy from the background shear"* [12, p. 6]).

2. **Two-Family Stress Cone and Intrinsic Force Generation:** These two wave families ($\sigma = \pm 1$) form two linearly independent covariance vectors ($v_1, v_2 \in \mathbb{R}^2$) at every point on the ring, and provide, through quadratic velocity products, the intrinsic stress ($T$) sustaining the vortex [12, p. 10]:

$$T_{\text{ring}} = c_1 v_1 + c_2 v_2, \qquad c_1, c_2 > 0 \tag{3}$$

   This intrinsic stress exactly balances the singular momentum deficit in the background through the nonlinear transport term $(u \cdot \nabla)u$, enabling the self-feeding of the vortex [12, pp. 5–6].

3. **Integer ($N$) Phase Encoding:** As proved in **Proposition C.2** [12, p. 10] of the article, a logarithmic phase ($\Phi_{\text{ring}} = N \ln X$) encoded with a large integer $N$ is loaded onto radial oscillations to satisfy this stress cone condition in the annulus region.

## 3.2 Energy Transfer Equation in the Black Hole Vortex: Fuel Flow from String to Vortex

Now let us apply this proved mechanism of the Navier–Stokes solution to the string falling onto the black hole event horizon, via the *Fluid–Gravity Correspondence* [7, 11].

The specific vibration energy of the $n$-th harmonic mode of a string freely falling in the $r \gg r_s$ region is initially $E_{\text{string},n}(0) = \hbar\omega_{n,0}$. When the string enters the event horizon annulus of the black hole, its frequency and specific energy in proper time drop toward zero, as derived in the third article [3]:

$$E_{\text{string},n}(\tau) = \hbar\omega_n(\tau) = \hbar\omega_{n,0}\, e^{-\Phi(\tau)} = \hbar\omega_{n,0}\, \tau^{+h} \xrightarrow{\tau\to 0} 0 \tag{4}$$

The covariant conservation of the total energy-momentum tensor in general relativity ($\nabla_\mu \left(T^{\mu\nu}_{\text{string}} + T^{\mu\nu}_{\text{vortex}}\right) = 0$) is an absolute law. Hence the inner specific energy lost by the string during stripping ($-\frac{dE_{\text{string}}}{d\tau} > 0$) cannot vanish; it is transferred **directly into the rotational and shear energy of the black hole vortex (into Vortex Fuel) through the stripped Light Dimension ($\mathcal{L}_{\text{light}}$)!**

The power pumped per unit proper time from the string, stripped into the black hole vortex as fuel ($\mathcal{P}_{\text{fuel}}$), is directly obtained from the proper-time derivative of Equation (4):

$$\mathcal{P}_{\text{fuel}}(\tau) = \frac{dE_{\text{vortex}}}{d\tau} = -\frac{d}{d\tau}\left(\sum_{n=1}^{\infty} E_{\text{string},n}(\tau)\right) = \sum_{n=1}^{\infty} \hbar\,\omega_{n,0}\,\Lambda(\tau)\, e^{-\Phi(\tau)} \tag{5}$$

Substituting the self-similar vortex scaling $\Phi(\tau) = h\ln(1/\tau)$ and $\Lambda(\tau) = \frac{h}{\tau}$, the instantaneous fuel power transferred to the vortex takes the following explicit form:

$$\mathcal{P}_{\text{fuel}}(\tau) = h\,\tau^{h-1} \sum_{n=1}^{\infty} \hbar\omega_{n,0} = \left(\frac{h\hbar c}{\ell_0} \sum_{n=1}^{\infty} n\, N_n\right) \tau^{h-1} \xrightarrow{\tau\to 0} \infty \tag{6}$$

The physical reality revealed by Equation (6) is striking:

- Since $0 < h < \frac{1}{100}$, at the singularity limit $\tau \to 0$, $\tau^{h-1} = \tau^{-(1-h)} \to \infty$; that is, at the exact moment of stripping, the instantaneous power transferred from the string to the vortex (the *fission pulse*) diverges to infinity!

- In contrast, when the total integral of this fuel power from $\tau = 1$ to $\tau = 0$ is taken, the **total fuel energy** transferred to the vortex exactly equals the initial specific energy of the string:

$$\begin{aligned} E_{\text{fuel, total}} &= \int_0^1 \mathcal{P}_{\text{fuel}}(\tau)\, d\tau = \sum_{n=1}^{\infty} \hbar\omega_{n,0} \int_0^1 h\,\tau^{h-1}\, d\tau \\ &= \sum_{n=1}^{\infty} \hbar\omega_{n,0} \left[\tau^h\right]_0^1 = E_{\text{string}}(0) \end{aligned} \tag{7}$$

Thus the entire specific energy of the string ($\hbar\omega_{n,0}$) is transferred as **fuel** to the vortex at the event horizon with no loss. Moreover, the right-moving ($\sigma - \tau$) and left-moving ($\sigma + \tau$) light modes in the Polyakov string correspond one-to-one to the two annulus pulse families ($\sigma = +1$ and $\sigma = -1$, Equation 3 [12, pp. 5, 10]) in the Navier–Stokes solution, feeding the stress cone of the vortex ($T_{\text{ring}} = c_1 v_1 + c_2 v_2$).

# 4 Resolution of the Black Hole Information Paradox: Spectral Fingerprint Engraved in Fuel

## 4.1 The Impasse of the Information Paradox

Since Stephen Hawking showed in 1974 [6] that black holes evaporate by radiating through quantum effects, modern physics faces the **Information Paradox**: if the information of matter falling into the black hole is trapped inside, and the Hawking radiation emitted outside is completely random (thermal/informationless), then when the black hole evaporates the information has vanished; this destroys **Unitarity (the Conservation of Information)**, the fundamental pillar of quantum mechanics.

## 4.2 The Natural Resolution with "Vortex Fuel"

*Stationary Strings Theorem IV* resolves this paradox in two arms through the "Cosmic Fission and Vortex Fuel" mechanism:

1. **Intrinsic Conservation (Stationary String Core):** As long as the black hole exists, the actual information of matter is preserved without degradation, with zero entropy, in the topological mode numbers ($\mathcal{I}_n = \omega_n \cdot \mathcal{L}_{\text{light}} = nc$) of the **Stationary Strings** that have passed into the $\omega_n = 0$ phase below the event horizon [1, 2, 3].

2. **Extrinsic Transfer (Encoded Vortex Fuel):** Noting Equations (6) and (7), the light shell stripped from the string and joining the event horizon vortex as "fuel" is **not** a random pile of heat! Every energy packet transferred to the vortex is modulated one-to-one with the stripped string's $n$ **topological quantum number** ($\Delta E_n = \frac{\hbar c}{\ell_0} n$) and with the logarithmic phase factor ($\Phi_n(\tau) = nh \ln(1/\tau)$, cf. the $N \log X$ phase in Proposition C.2 [12, p. 10]) in the Navier–Stokes annulus solution.

Hence the angular momentum and energy spectrum $S_{\text{fuel}}(\omega)$ of the vortex fuel at the event horizon carries a complete **holographic imprint (spectral fingerprint)** of the $\{n_k\}$ information sequence of the infalling strings:

$$\mathcal{T}_{\mu\nu}^{(\text{fuel})} = \sum_{n=1}^{\infty} n \, (c_{1,n} v_{1,\mu\nu} + c_{2,n} v_{2,\mu\nu}) \, e^{in\Phi(\tau)} \tag{8}$$

As the black hole emits Hawking radiation or gravitational waves into outer space, it spends energy from this **Light Fuel Ring** ($\mathcal{T}_{\mu\nu}^{(\mathbf{fuel})}$) at the event horizon. Since the fuel itself is encoded with the $n$ quantum numbers, the phase correlations of the emitted photons and gravitons carry this $n$ information back to the outer universe completely. Information is never lost at any moment, inside or outside; unitarity is absolutely conserved.

# 5 Elimination of the $r = 0$ Singularity at the Center: Singularity-Free "Pure Information Core"

## 5.1 The Classical Singularity Problem

The Schwarzschild metric in classical general relativity:

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right)c^2dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1}dr^2 + r^2d\Omega^2 \tag{9}$$

leads the spacetime curvature (Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48G^2M^2}{c^4r^6}$) at the exact center ($r \to 0$) of the black hole to diverge to infinity ($\infty$). The reason is the assumption in classical physics that matter continues to remain normal "matter/energy" after crossing the event horizon and collapses to a single point.

## 5.2 The Equation of State of Stationary Strings ($\omega_n = 0$) and Solution of the Einstein Field Equations

According to the *Stationary Strings Theorem*, however, when matter passes through the shear of the vortex at the event horizon, its kinetic light/energy skin is stripped and the vibration of the inner strings is completely nulled ($\omega_n = 0$) [1, 2, 3]!

The energy-momentum tensor $T_\nu^\mu = \mathrm{diag}(-\rho c^2, P_r, P_\perp, P_\perp)$ of a string network passing below the event horizon, when derived from the Polyakov action integral, consists of two components:

1. **Kinetic Vibration Pressure ($P_{\mathbf{kinetic}}$):** The positive pressure born from the time-dependent oscillations ($\dot{X}^2 \propto \omega_n^2$) of the string. Since $\omega_n \to 0$ as a result of stripping, this term **vanishes completely** below the horizon:

$$P_{\text{kinetic}} \propto \sum_{n=1}^{\infty} \omega_n^2(\tau)\, q_n^2 \xrightarrow{\omega_n = 0} 0 \tag{10}$$

2. **Topological Information Tension ($P_{\mathbf{information}}$):** When vibration halts ($\omega_n = 0$), only the topological information invariant ($\mathcal{I}_n = nc$) of Equation (1) and the spatial tension fabric ($\partial_\sigma X$) of the stationary string remain. The radial equation of state of a

stationary string condensate with zeroed kinetic energy, possessing only topological winding information ($n$), produces a **negative pressure (vacuum-information tension)** equal in magnitude to its energy:

$$P_r = -\rho_{\text{information}}(r)\, c^2 \tag{11}$$

The core information density distribution formed by stationary strings with total topological information quantum $N_{\text{information}} = \sum_k n_k$, under the quantum-information capacity bound, is of the form:

$$\begin{aligned} \rho_{\text{information}}(r) &= \frac{3M}{\pi} \frac{r_{\text{information}}^3}{\left(r^3 + r_{\text{information}}^3\right)^2}, \\ r_{\text{information}} &= \left(\frac{3G\hbar N_{\text{information}}}{c^3}\right)^{1/3} = \ell_{\text{Planck}} \left(3N_{\text{information}}\right)^{1/3} \end{aligned} \tag{12}$$

Here $r_{\text{information}}$ is the characteristic radius of the **Pure Information Core** formed by the stationary strings.

Now let us insert Equations (11) and (12) directly into the $(t,t)$ component of the **Einstein Field Equations** ($G_\nu^\mu = \frac{8\pi G}{c^4} T_\nu^\mu$) and integrate from $r = 0$ to $r$:

$$\frac{d}{dr}\left[r\left(1 - f(r)\right)\right] = \frac{8\pi G}{c^2} r^2 \rho_{\text{information}}(r) \tag{13}$$

When this integral is solved exactly, the **Singularity-Free Black Hole Metric of the Stationary Strings Theorem** is obtained:

$$ds^2 = -f(r)c^2dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \qquad f(r) = 1 - \frac{2GMr^2}{c^2\left(r^3 + r_{\text{information}}^3\right)} \tag{14}$$

The two great mathematical proofs provided by Equation (14) are:

1. **Full Agreement with Einstein Relativity in Outer Space ($r \gg r_{\text{information}}$):** Since $r^3 + r_{\text{information}}^3 \approx r^3$ in all regions outside the information core, the metric function reduces to the classical Schwarzschild solution as $f(r) \approx 1 - \frac{2GM}{rc^2}$.

2. **Complete Disappearance of the Singularity at the Center ($r \to 0$):** Descending to the exact center of the black hole ($r \to 0$), the $r_{\text{information}}^3 \propto N_{\text{information}}$ (Topological Information Volume of the Stationary Strings) in the denominator prevents nulling, and the metric function attains a smooth **De Sitter Information Core** behavior:

$$\lim_{r\to 0} f(r) = 1 - \left(\frac{2GM}{c^2 r_{\text{information}}^3}\right) r^2 = 1 - \frac{r^2}{L_{\text{eff}}^2} \longrightarrow 1 \quad \text{(No Singularity!)} \tag{15}$$

Computing the Kretschmann spacetime curvature scalar at the center yields not infinity ($\infty$) but a completely **finite and regular** value:

$$\begin{aligned}\lim_{r\to 0} K(r) &= \lim_{r\to 0} R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \\ &= \frac{96G^2M^2}{c^4 r_{\text{information}}^6} = \frac{32c^2M^2}{3\hbar^2 N_{\text{information}}^2} < \infty\end{aligned} \tag{16}$$

It is thus proved that: **at the center of black holes there is no infinite-density singularity ($r = 0$) where the laws of physics collapse.** The stationary strings ($\omega_n = 0$) stripped of their energy shell in the vortex form a zero-entropy, finite-curvature, perfectly ordered **Pure Information Core** at the center, halting the collapse of spacetime.

# 6 Observational Proof and Experimental Verification Setup

A theoretical model gains scientific certainty by being tested through the numerical and observational predictions it produces. *Stationary Strings Theorem IV* predicts two concrete experimental proofs testable both at astrophysical observatories (LIGO/Virgo/KAGRA and the future LISA) and in fluid/quantum laboratories.

## 6.1 Astrophysical Proof: Gravitational Wave Echoes

In classical general relativity, since the black hole event horizon is assumed to be a completely empty one-way membrane, the gravitational wave signal (*ringdown*) produced when two black holes collide damps rapidly and ends.

In our model, however, the black hole consists of two physical layers: the outer **Vortex Fuel Ring ($r \approx r_s$)** and the inner **Stationary String Information Core ($r = r_{\text{information}}$)**. Gravitational waves produced during the collision must reflect between the angular momentum barrier and the inner Stationary String Information Core, producing periodic **Quantum Echoes** after the main collision signal.

The round-trip time of the wave between these two layers ($\Delta t_{\text{echo}}$), computed via the tortoise coordinate $\Delta r^* = \int \frac{dr}{f(r)}$ over the metric of Equation (14), is directly determined by the total vortex tension $\Phi_* = \ln(r_s/r_{\text{information}})$:

$$\Delta t_{\text{echo}} = \frac{2}{c}\int_{r_{\text{information}}+\delta}^{r_{\text{barrier}}} \frac{dr}{f(r)} \approx \frac{4GM}{c^3}\,\Phi_* = \frac{4GM}{c^3}\ln\left(\frac{r_s}{\ell_{\text{Planck}}(3N_{\text{information}})^{1/3}}\right) \tag{17}$$

The frequency spectrum of these echoes, in turn, is quantized directly with the topological mode number $n$ ($\mathcal{I}_n = nc$) of the stationary strings:

$$f_{\text{echo},n} = \frac{n}{2\,\Delta t_{\text{echo}}} = \frac{nc^3}{8GM\Phi_*} \tag{18}$$

This equation is a precise and falsifiable mathematical prediction searchable in the damping (post-merger ringdown) data of black hole mergers.

## 6.2 Laboratory Experiment Setup: Acoustic and Superfluid Vortex Singularity Experiment

Since the $\mathcal{I}_n = \omega_n(\tau) \cdot \mathcal{L}(\tau) = nc$ law of Equation (1) and the vortex fuel transfer of Equation (5) rest on the scale-invariant hyperbolic symmetry of the Navier–Stokes and Euler equations, they can be directly tested with an **Analog Black Hole Vortex Setup** to be built in the laboratory:

1. **Acoustic Event Horizon Vortex Tank:** A low-viscosity fluid (or superfluid Helium-4 / Bose–Einstein Condensate) is rotated to form an axisymmetric vortex toward a central drain. The radius where the radial flow velocity $v_r(r)$ equals the wave propagation speed $c_s$ in the medium ($v_r(r_H) = c_s$) is mathematically identical to the black hole event horizon metric.

2. **Sending the String-Wave Ring:** From the outer region of the vortex, a closed **elastic/vortex wave ring** (or Kelvin wave on a quantum vortex ring) carrying $n$ standing wave nodes (harmonic modes) and vibrating at frequency $\omega_{n,0}$ is released toward the vortex center.

3. **Three Experimental Proofs to Measure:** With high-speed laser interferometry and Particle Image Velocimetry (PIV), as the ring enters the $r \to r_H$ boundary (annulus region), the following three phenomena are measured simultaneously:

   - rapid stretching of the tangential/axial length of the ring by the $\mathcal{L}(\tau) \asymp \tau^{-h}$ law,
   - damping-to-zero ($\omega_n \to 0$) of the inner vibration frequency on the ring by the $\omega_n(\tau) \asymp \tau^{+h}$ law, and the lost energy producing secondary oscillatory pulses ($\sigma = \pm 1$) in the ring region accelerating the main vortex (Vortex Fuel proof),
   - verification that throughout the process the product $\omega_n(\tau) \cdot \mathcal{L}(\tau) = nc_s$ and the number $n$ of nodes (topological information) on the ring stay constant!

# 7 Philosophical Interpretation and the Integrity of the Tetralogy

With this fourth article, the *Stationary Strings Theorem* attains a complete cosmological and mathematical architecture extending from the beginning of existence to the heart of black holes:

- **Theorem I (Ontological Foundation [1]):** Time and entropy are born with the vibration ($\omega > 0$) of strings; when vibration halts ($\omega = 0$), time and chaos are erased and absolute order (Pure Information) begins.
- **Theorem II (Relativity Bridge [2]):** Gravitational time dilation and string frequency obey the same law ($t' = k_t \cdot f_0\sqrt{1 - \frac{2GM}{rc^2}}$); black holes are phase transducers converting energy into information.
- **Theorem III (Two-Dimensional Duality and Ontological Stripping [3]):** Matter consists of two inversely proportional dimensions, **String (Frequency)** and **Light (Flow)**. In the black hole vortex, per the Navier–Stokes vortex singularity [12] and Polyakov action, the light dimension is pulled to infinity ($\mathcal{L}_{\text{light}} \to \infty$) while the string frequency drops to zero ($\omega_n \to 0$) in proper time; these two opposite forces dissect the fabric of matter like flaying skin, separating it into the topological information invariant ($\mathcal{I}_n = nc$).
- **Theorem IV (Cosmic Fission, Vortex Fuel and Singularity-Free Core):** This stripping is an obligatory dimensional fission like the fragmentation of the atom. The stripped light and specific-energy dimension turns into **Vortex Fuel** ($\mathcal{P}_{\text{fuel}}$) feeding the vortex at the event horizon ring and carrying the $n$ information to the outer universe, resolving the Information Paradox; the incoming **Stationary String** establishes a negative-pressure ($P_r = -\rho_{\text{information}}c^2$) **Pure Information Core** at the center, eliminating the $r = 0$ singularity.

No energy and no information is lost in the universe: matter is stripped in the vortex; while its shadow, energy, becomes fuel to the vortex, its essence, information, awaits rebirth in infinite order at the heart of the black hole.

> *"Without fragmentation of matter, information cannot be liberated;*
> *while the stripped light becomes fire to the vortex, the stationary string withdraws into the infinite tranquility of the center."*
> — Fahri Meral, Stationary Strings Theorem IV (2026)

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